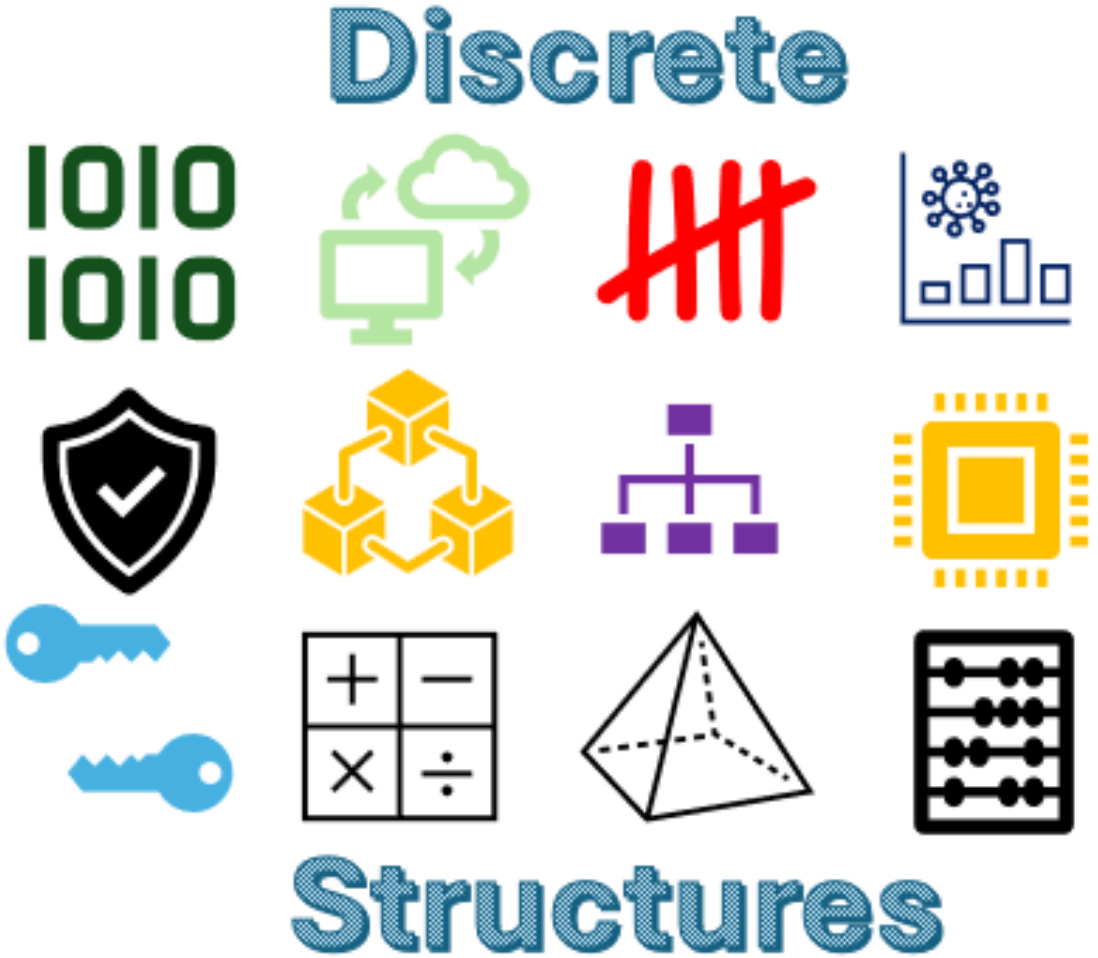
COMP 2310: Discrete Structures
Fall 2026

Chapter 2: Logic



Conditional statement

- Denoted by ⇒
- p ⇒ q, represents: if p is true then q where p and q are propositions
- p = hypothesis, q = conclusion
- p ⇒ q reads: if p is true then q is true (p implies q).
- q is not exclusively depended on p or p does not always cause q
  - p = if it is raining, q = the ground is wet (not the only reason)
- **Converse (reversing)** of p ⇒ q is q ⇒ p
- **Contrapositive (logically equivalent)** of p ⇒ q is ~p ⇒ ~q

TABLE 2.4

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Exercise

- Make a truth table for $p \Rightarrow q \Rightarrow r$
- $(p \Rightarrow q) \Rightarrow r$
  - If $(p \Rightarrow q)$ is true then r is true
- $p \Rightarrow (q \Rightarrow r)$
  - If p is true then $(q \Rightarrow r)$ true

Biconditional statement

- $p \Leftrightarrow q$ reads as: p if and only if q
  - p: if it rains, q: I get wet

TABLE 2.5

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Types of statements

- **Tautology:** A statement that is true for all possible values of its propositional variables is called a tautology (*it will snow tomorrow or it will not snow tomorrow*)
  - [p or ¬p]
  - p⇔q is a tautology if it is true no matter what truth values p and q take.
- **Logical equivalence:**
  - p⇔q is a tautology, it means that p and q are logically equivalent. This means p and q have the same truth value in every possible scenario.
    - whenever it is raining, the ground is wet, and whenever the ground is wet, it is raining. Therefore, p≡q
  - **Denoted by** ≡

Examples

p	q	$p \vee q$	$q \vee p$	$(p \vee q) \Leftrightarrow (q \vee p)$
T	T			
T	F			
F	T			
F	F			

p	q	$p \Rightarrow q$	$\sim p$	$(\sim p) \vee q$
T	T			
T	F			
F	T			
F	F			

Example

- Let S = {2, 3, 5} and let P(n) = n – 1 is even, Q(n) = n + 3 is odd and n∈s.

  P(2) ⇒Q(2) is ?

  P(3) ⇒ Q(3) is ?

  P(2) ⇒Q(2) is True

  P(3) ⇒ Q(3) is False

- Review theorem 1 in Chapter 2

Methods of proof

- Types of proof:
  - Direct proof: using definition or established theorem
  - Proof by cases: by examining all possible truth value situation
- $(p1 \wedge p2 \wedge \cdots \wedge pn) \Rightarrow q$ (1)
  - p: hypothesis/premise
  - q: conclusion
  - To prove the (1), we need to prove the implication is tautology
  - q will be true if all p is true

Mathematical induction

- A powerful proof technique
- Used to prove statements for all integers greater than or equal to some fixed number
- Steps
  - Basis step
  - Induction step
- Using notations:
  - A method for proving statements of the form $\forall n \geq n_0\ P(n)$
  - Basis step: Show $P(n_0)$ is true
  - Induction step: Show $P(k) \Rightarrow P(k+1)$ is true for $k \geq n_0$

Mathematical induction

- **Basis Step**: Prove $P(n_0)$ directly.
- **Induction Step**: Assume P(k) is true, and show it implies P(k+1) is true
- By following these steps, you prove that P(n) holds for all $n \geq n_0$

Mathematical induction: example

- Sum of natural numbers

Example 1 Show, by mathematical induction, that for all $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Solution

Let P(n) be the predicate $1+2+3+\cdots+n = \frac{n(n+1)}{2}$. In this example, $n_0 = 1$.

Basis Step

We must first show that P(1) is true. P(1) is the statement

$$1 = \frac{1(1+1)}{2},$$

which is clearly true.

Induction Step

We must now show that for $k \geq 1$, if P(k) is true, then P($k + 1$) must also be true. We assume that for some fixed $k \geq 1$,

$$1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}. \tag{1}$$

We now wish to show the truth of P($k + 1$):

$$1 + 2 + 3 + \cdots + (k+1) = \frac{(k+1)((k+1)+1)}{2}.$$

The left-hand side of P($k + 1$) can be written as $1 + 2 + 3 + \cdots + k + (k + 1)$ and we have

$$\begin{aligned}
&(1 + 2 + 3 + \cdots + k) + (k+1) \\
&\quad = \frac{k(k+1)}{2} + (k+1) && \text{using (1) to replace } 1 + 2 + \cdots + k \\
&\quad = (k+1)\left[\frac{k}{2} + 1\right] && \text{factoring} \\
&\quad = \frac{(k+1)(k+2)}{2} \\
&\quad = \frac{(k+1)((k+1)+1)}{2} && \text{the right-hand side of P}(k+1)
\end{aligned}$$

Thus, we have shown the left-hand side of P($k + 1$) equals the right-hand side of P($k + 1$). By the principle of mathematical induction, it follows that P(n) is true for all $n \geq 1$. ◆

Mathematical induction: example

- Calculating the square of a given number A
- Input (A) must be a positive integer

```
FUNCTION SQ(A)
1. C ← 0
2. D ← 0
3. WHILE (D ≠ A)
     a. C ← C + A
     b. D ← D + 1
4. RETURN (C)
END OF FUNCTION SQ
```

Basis Step

P(0) is the statement $C_0 = A \times D_0$, which is true since the value of both C and D is zero "after" zero passes through the **WHILE** loop.

Induction Step

We must now use

$$\mathrm{P}(k)\colon C_k = A \times D_k \tag{2}$$

to show that $\mathrm{P}(k+1)\colon C_{k+1} = A \times D_{k+1}$. After a pass through the loop, C is increased by A, and D is increased by 1, so $C_{k+1} = C_k + A$ and $D_{k+1} = D_k + 1$.

left-hand side of $\mathrm{P}(k+1)$: $C_{k+1} = C_k + A$

$= A \times D_k + A$ using (2) to replace C_k

$= A \times (D_k + 1)$ factoring

$= A \times D_{k+1}$ right-hand side of $\mathrm{P}(k+1)$

By the principle of mathematical induction, it follows that as long as looping occurs, $C_n = A \times D_n$. The loop must terminate. (Why?) When the loop terminates, $D = A$, so $C = A \times A$, or A^2, and this is the value returned by the function SQ.